

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech. (EE) Examination****December 2016****EE307 Electrical Power System III****Date: 01/12/2016, Thursday****Time: 01.30 p.m. to 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1** Choose the correct option. **[07]**

- I. The arc voltage in a circuit breaker is
 - (a) in the phase with the arc current
 - (b) lagging the arc current by 90°
 - (c) leading the arc current by 90°
 - (d) in phase with system voltage
- II. In a circuit breaker, the active recovery voltage depends upon
 - (a) circuit conditions
 - (b) power factor
 - (c) armature reaction
 - (d) all of these
- III. Which of the following sequences of operations represents the rated operating duty cycle of a circuit breaker? (O-open; C-close; t = 3s; T = 3 minutes)
 - (a) O-t-CO-T-CO
 - (b) O-CO-t-CO-T-C
 - (c) O-C-t-OC-T
 - (d) O-CO-T-CO-T-C
- IV. For a high speed circuit breaker the total clearing time is around
 - (a) few minutes
 - (b) few seconds
 - (c) 1 to 2 cycles
 - (d) 5 to 20 cycles
- V. The normal frequency _____ voltage that appears across the circuit breaker poles after final arc extinction has occurred is termed as the recovery voltage.
 - (a) average
 - (b) rms
 - (c) peak
 - (d) transient
- VI. When the restriking voltage is at its peak value, the value of RRRV is _____.
 - (a) zero
 - (b) maximum
 - (c) half the maximum value of RRRV
 - (d) none of these.
- VII. The resistance of an electric arc can be increased by
 - (a) increasing the concentration of ionized particles
 - (b) reducing the arc length
 - (c) splitting the arc
 - (d) increasing the arc x-section

Q – 2. Answer any TWO.

- A** Derive the expression for restriking voltage in terms of system capacitance and inductance. Also find the condition when the restriking voltage is maximum and RRRV is maximum. [07]
- B** With neat diagram explain the phenomena of current chopping. [07]
- C** A 50 Hz three phase alternator with grounded neutral has an inductance of 2.5 mH per phase and is connected to the busbar through a circuit breaker. The capacitance to earth of the circuit between the alternator and the circuit breaker is 0.003 μ F per phase. Due to a short on the busbars the breaker opens when the rms value of current is 10000 A. Determine the following. [07]
- Frequency of oscillations
 - Active recovery voltage
 - Time for maximum RRRV
 - Maximum RRRV.

Q - 3 Answer any TWO

- A** Explain the physical and dielectric properties of SF₆ gas. [07]
- B** Explain different methods for arc extinction. [07]
- C** With neat and clean figures, explain the construction and working of Air Blast Circuit Breaker. [07]

SECTION – II**Q - 4 Choose the correct option. [07]**

- If the KVA base of a system is halved then what will be the change in its per unit impedance?
(a) Doubled (b) halved (c) 4 times (d) none of these
- The zero sequence current of a generator for line to ground fault is $j2.4$ pu. Then the current through the neutral during the fault is
(a) $j2.4$ pu (b) $j0.8$ pu (c) $j7.2$ pu (d) $j0.24$ pu
- Consider the following statements regarding the fault analysis:
 - The neutral grounding impedance Z_n appears as $3Z_n$ in zero-sequence equivalent circuit.
 - For faults on transmission lines, 3-phase fault is the least severe amongst other faults.
 - The positive and negative sequence networks are not affected by method of neutral grounding.
 Which of the statements given above are correct?
(a) 2 and 3 (b) 1 and 2 (c) 1 and 3 (d) 1, 2 and 3
- A fault occurring at the terminals of an unloaded synchronous generator operating at its rated voltage has resulted in the following values of currents and voltages

$$I_{a0}=j2.37 \text{ pu}, \quad I_{a1}=-j3.05 \text{ pu}, \quad I_{a2}=j0.68 \text{ pu},$$

$$V_{a0} = V_{a1} = V_{a2} = 0.237 \text{ pu}$$

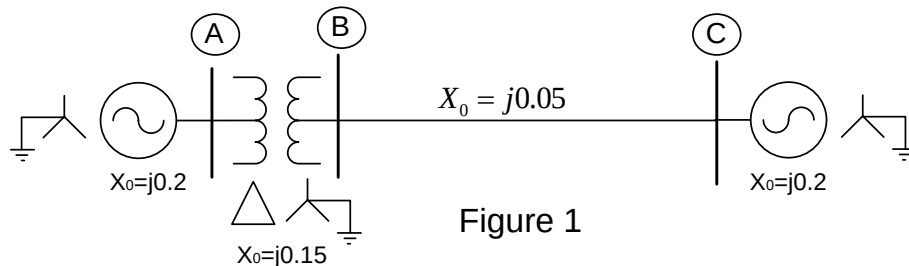
Which one of the following faults has occurred?

- (a) L-L fault. (b) L-G fault. (c) L-L-G fault. (d) L-L-L fault.

V. If the positive-, negative- and zero-sequence reactances of an element of a power system are 0.3, 0.3 and 0.8 pu respectively, then the element would be a

- (a) synchronous generator (b) synchronous motor
(c) static load (d) transmission line

VI. For the network shown in figure 1, the zero sequence reactances in p.u. are indicated. The thevenins equivalent zero sequence reactance for fault at bus C is



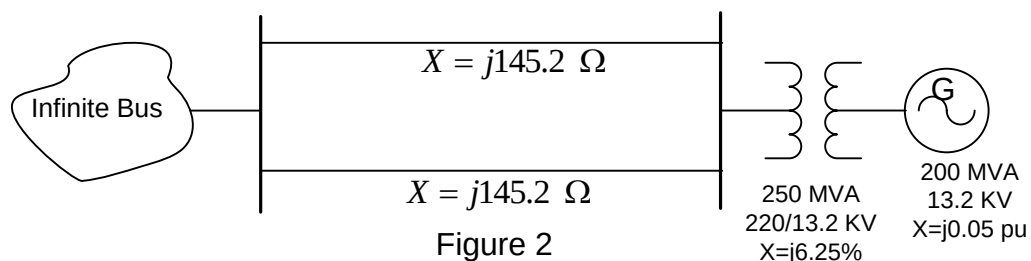
- (a) 0.12 (b) 0.30 (c) 0.10 (d) 0.20

VII. A single line to ground fault occurs on a three-phase isolated neutral system with a line to neutral voltage of V kV. The potential on the healthy phase rise to a value equal to

- (a) $\sqrt{2}V$ kV (b) $\sqrt{3}V$ kV (c) $3V$ kV (d) $(1/\sqrt{3})V$ kV

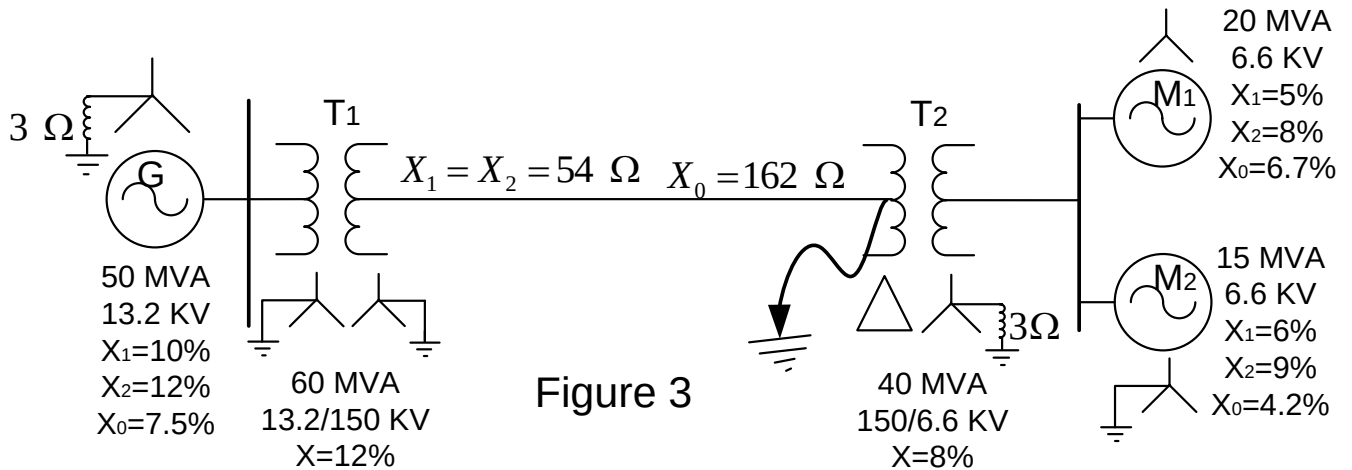
Q – 5. Answer any TWO

- A** Explain the advantages of per-unit system and derive the fundamental relations between the actual values and per-unit values on chosen base values for a three phase system. [07]
- B** In connection with sequence components, prove the power invariance theorem. [07]
- C** A 13.2 kV, 200 MVA generator is connected to infinite bus as shown in figure 2. [07]
If three phase bolted fault occurs at middle of any transmission line, find the total fault current. Also find the contribution of fault current from infinite bus and from generator shown in figure. (Take base as per generator rating)



Q – 6. Answer any TWO

- A** Using appropriate interconnection of sequence networks, derive the equation for a [07]
single line to ground fault in a power system with a fault impedance of Z_f .
- B** A part of power system is shown in figure 3. Find the value of LL fault current in [07]
ampere when fault occurs at the High voltage terminal side of transformer T_2 as
shown in figure. **Take base as per generator rating.**



- C** A power system is shown in figure 4. Draw the zero sequence reactance diagram [07]
taking base as per generator rating. The zero sequence impedance of transmission
line is three times that of negative sequence impedance.

